PROJECT PORTFOLIO 02

District-level Inequality & Competitiveness

Theil Index and Regional Competitiveness Index | Northern Bangladesh

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Project overview

This project compares socio-economic inequality and regional competitiveness across 16 districts in northern Bangladesh. It uses the Theil Index to assess inequality across selected service and literacy indicators, and the Regional Competitiveness Index (RCI) to compare district-level competitiveness. The two measures are combined to create a four-category district typology.

Study area

Rajshahi, Chapainawabganj, Natore, Bogura, Joypurhat, Lalmonirhat, Gaibandha, Thakurgaon, Panchagarh, Sirajganj, Pabna, Naogaon, Rangpur, Dinajpur, Nilphamari and Kurigram.



Study area map reproduced from the original group report.

Objectives

- Measure district-level inequality using the Theil Index.
- Compare regional competitiveness using the RCI.
- Classify districts by combining competitiveness and inequality to identify different planning priorities.

Data and tools

- Theil Index variables: education centres, healthcare centres, banks, haat/bazars and literacy rate.
- RCI variables stated in the report: total population, education centres, healthcare centres and literacy rate.
- Sources named: BBS and BANBEIS. Tools: Microsoft Excel and QGIS.

Methods

Theil Index

The Theil Index is used to compare inequality across the selected indicators. The source report calculates indicator-level values and combines the five variables to produce a district-level inequality score. Lower values are interpreted as lower inequality, with zero representing perfect equality in the report's description.

$$\text{Theil } T = (1/N) \times \Sigma \left[(y_i / \bar{y}) \times \ln(y_i / \bar{y})\right]$$

The equation above is transcribed in a readable form from the report. The original calculation workbook should be consulted to verify the aggregation and implementation details.

Regional Competitiveness Index

The RCI is calculated as the average of standardized indicator values. The source describes Z-score standardization followed by normalization for comparison, using four indicators: population, education centres, healthcare centres and literacy rate.

RCI = average of the standardized indicator scores

Combined district classification

The report combines the inequality and competitiveness results to place districts into four analytical categories. This typology is used to compare district conditions and discuss differentiated planning priorities. It is an analytical classification developed for the study, not an official government classification.

Results and spatial outputs

The original report presents district-level indicator results, comparisons of inequality and competitiveness, and maps showing the resulting spatial patterns across the study area.

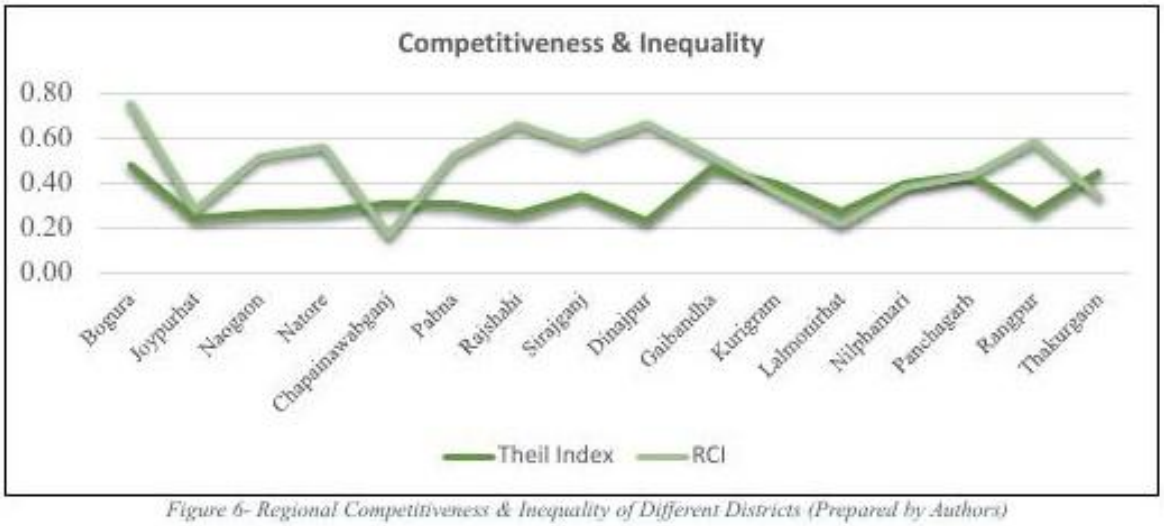
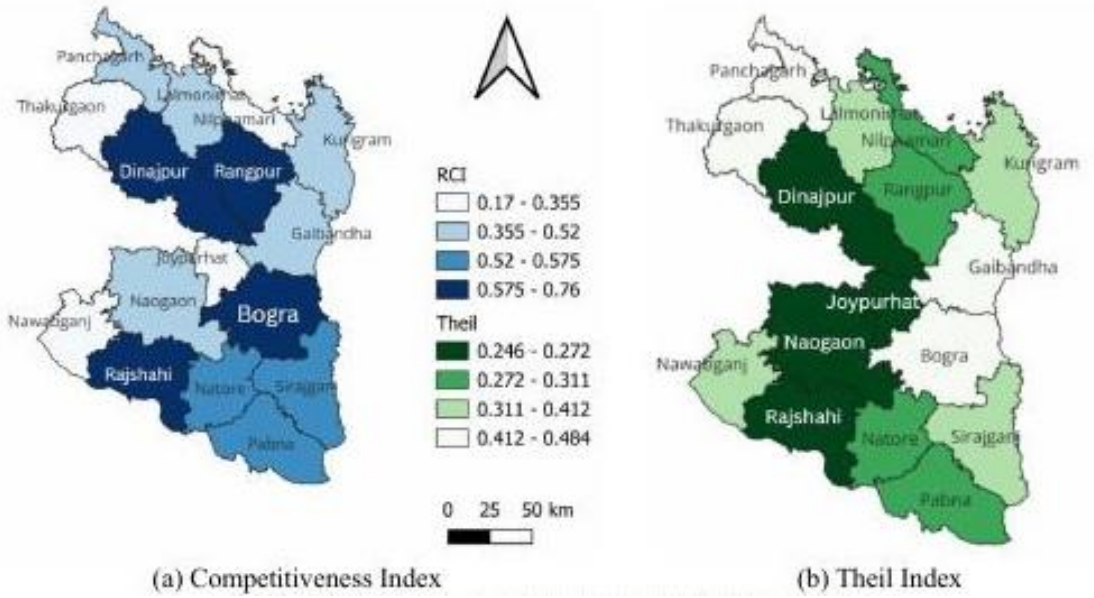


Figure 6- Regional Competitiveness & Inequality of Different Districts (Prepared by Authors)

By going through the figure 3 (a,b) we get to know that, the regional competitiveness index is higher in some areas which does not ensure equal distribution or growth of all the facilities those indicates the overall growth.



Charts and district-level spatial outputs reproduced from the original group report.

District typology and planning relevance

As per figure 3 we have seen that though Bogra is highly competitive in terms of gaining attraction from the investors, but also it says that, Bogra maintains a highly concerning inequality. Whereas, Rajshahi and Dinajpur is consistent in terms of both positive sides of highly competitive and low inequality. And we can make a quick summary based on this classification. The figure 4 helps us to understand overall image of the northern region. Where the greens are supposed to be good and whites are supposed to be less competitive but more unequally distribution of resources. In Bogura, we can see it blue, which means it is more competitive and more inequal allocation of resources. But Rajshahi is different here as well as Dinajpur. They are both competitive highly and also maintains the balanced growth of everything across sectors. Later on, we can classify them as per the mean competitiveness and Theil index. The next table 3 and figure 5 shows the classifications and criterion clearly. Here the reds areas are required the most priority which poses high inequality as well as low competitiveness. The yellow ones are great.



Figure 8: RCI and Theil Index Combined District Priority Map (Prepared by Authors)

Table 7- Classification of Districts (Prepared by Authors)

Category	Criteria	Districts
Core Growth Zone	High Competitiveness-Low Inequality	Rangpur, Dinajpur, Natore, Pabna, Rajshahi
Unequal Growth Zone	High Competitiveness-High Inequality	Bogura, Gaibandha, Shirajganj
Stable Hinterland	Low Competitiveness-Low Inequality (Critical Zone)	Joypurhat, Lalmonirhat, Chapainawabganj, Naogaon
Priority Zone	Low Competitiveness-High Inequality	Kurigram, Nilphamari, Panchagarh, Thakurgaon

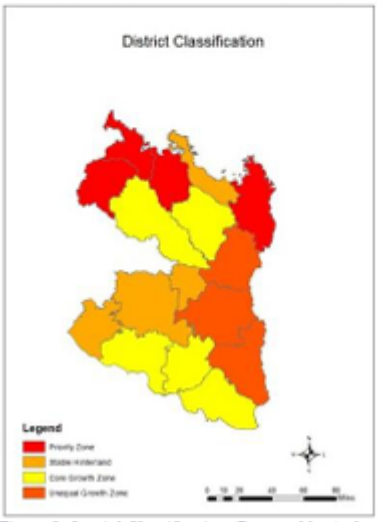


Figure 9- Spatial Classification (Prepared by Authors)

District typology and priority map reproduced from the original group report.

The combined typology offers a way to distinguish districts with different combinations of inequality and competitiveness. It can inform discussion of where service provision, access and regional development interventions may require different priorities.

Limitations and verification

The source report contains a contradiction between the written RCI ranking and the RCI values/ranks in its table. This portfolio does not repeat an unverified district ranking. The report does not document every normalization step or provide the original calculation workbook. The Theil Index aggregation and all district-level calculations should be checked against the source data before formal use.